

Advanced Geovisualisation

Module Introduction

ENVS456 – week 1
Gabriele Filomena



Agenda

- Module's aims & skills
- Geovisualisation
- Logistics

Aims

- Develop an understanding of geovisualisation principles and techniques, combining theoretical knowledge of map design with practical computational skills.
- Address geographic research problems, utilising the internet as both a source of geographical data and an analytical platform.
- Create effective visual narratives on human and physical geography-related issues.

Learning outcomes and Skills

1. Visualise and represent geo-data through static and dynamic maps.
2. Collect Web-based data.
3. Generate interactive maps and dashboards.
4. Understand basic concepts of spatial network analysis
5. Acquire introductory skills in satellite imagery, spatio-temporal, and mobility data visualisation.
6. Manipulate geo-data through scripting in Python.

Schedule

- *W2 : Map Design & Statistical Visualisation*
- *W3 : Interactive Maps*
- *W4 : Web Architectures & API*
- *W5 : Review & Assignment I deadline (26th February)*
- *W6: Retrieving Data From OpenStreetMap*
- *W7: From Pixels to Patterns: Visualising Satellite Image Classification*
- *W8: No teaching*
- *W9: Simulating agents with ABM (tbc)*
- *W10 : Time and Flows*
- *W11 : Review & Assignment II deadline (30th April)*

Assignment I:

Exploring APIs and Interactive Maps in Python. Jupyter Notebook

- ~250 w. introducing the research problem, the context, and existing research.
- Code: Introductory Static Maps (2), presenting the topic and the context.
- ~250 w. presenting and motivating the chosen data sources.
- Code: An API request written by your own - All the necessary steps for making your API work and for data cleaning/exploration.
- ~200 w. with your description of what your API is, how it works and how it will made your map possible.
- Code: An interactive final map. This should be fed with data obtained through the API request.
- ~200 w. description of how your interactive map works, its components and your design ideas.
- ~150 w. to summarise your research problem and how you tackled it by means of geovisualisation tools (Conclusion).

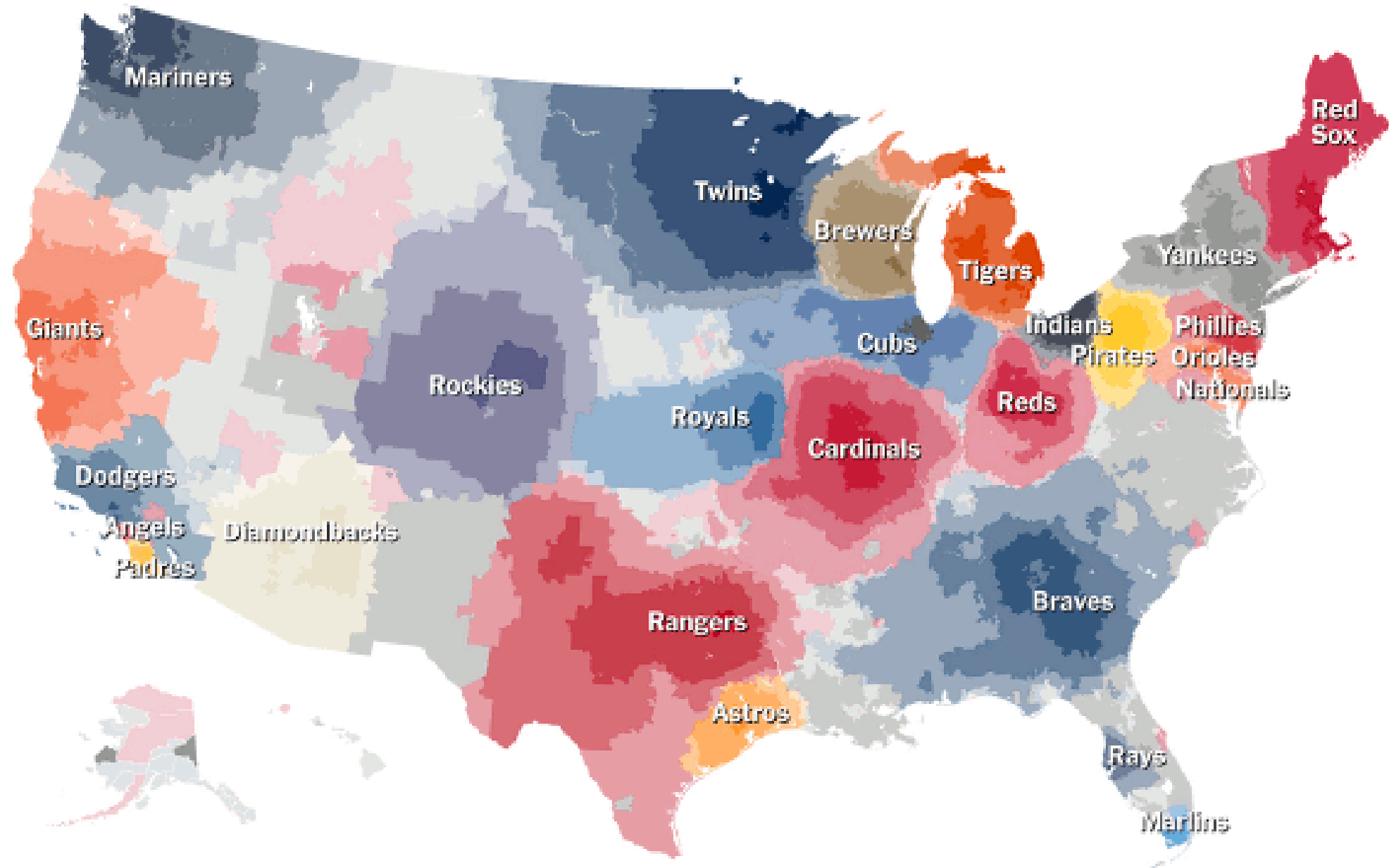
What is geovisualisation?

Geovisualisation

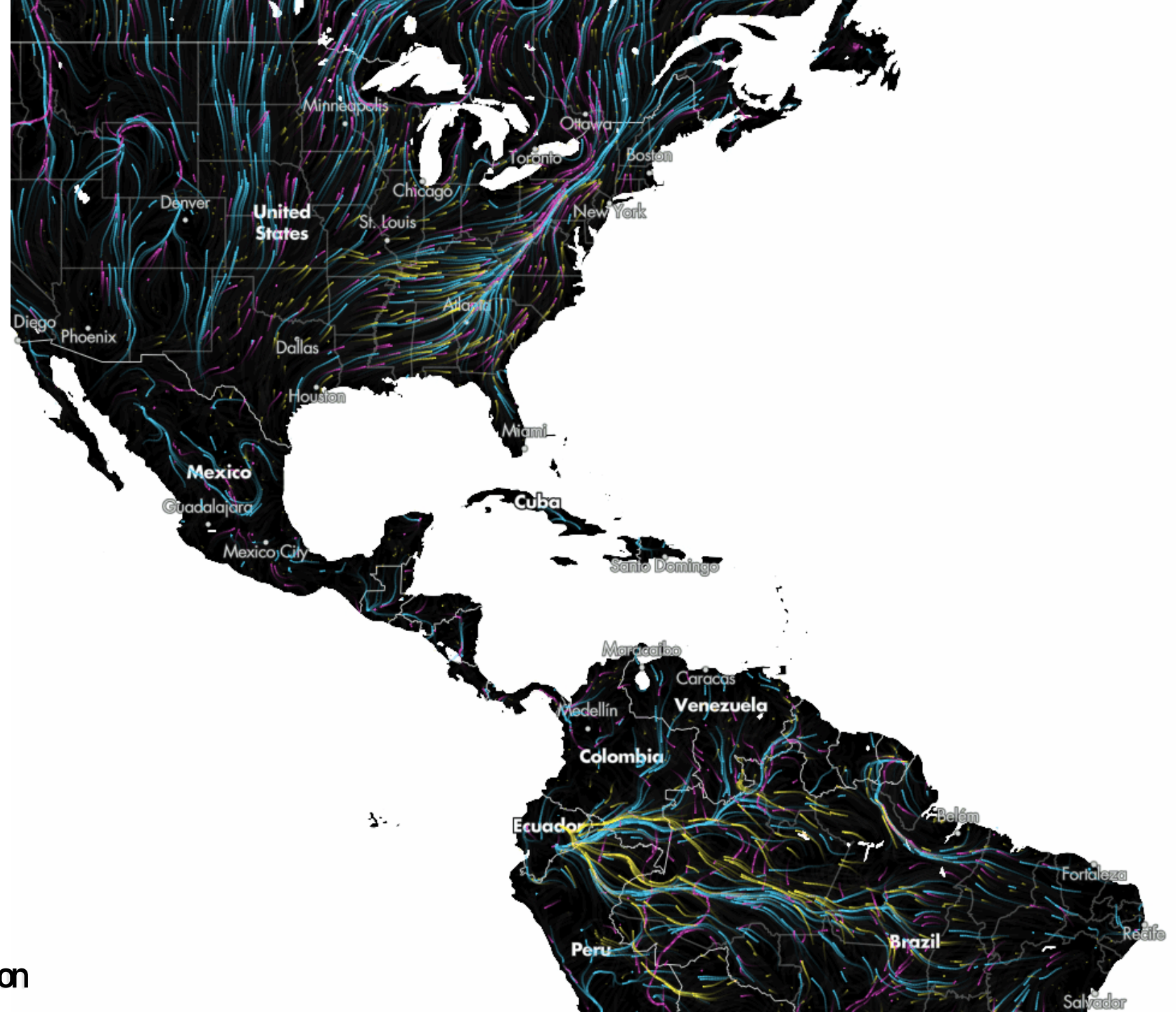
- It is the (interactive) exploration and communication of spatial data through maps and related visual displays.
- It integrates cartography, information visualisation, and analysis to help people identify patterns, relationships, and changes across space and time.

It supports both analytical reasoning (discovering insights) and effective presentation (explaining them to others).

Static Maps

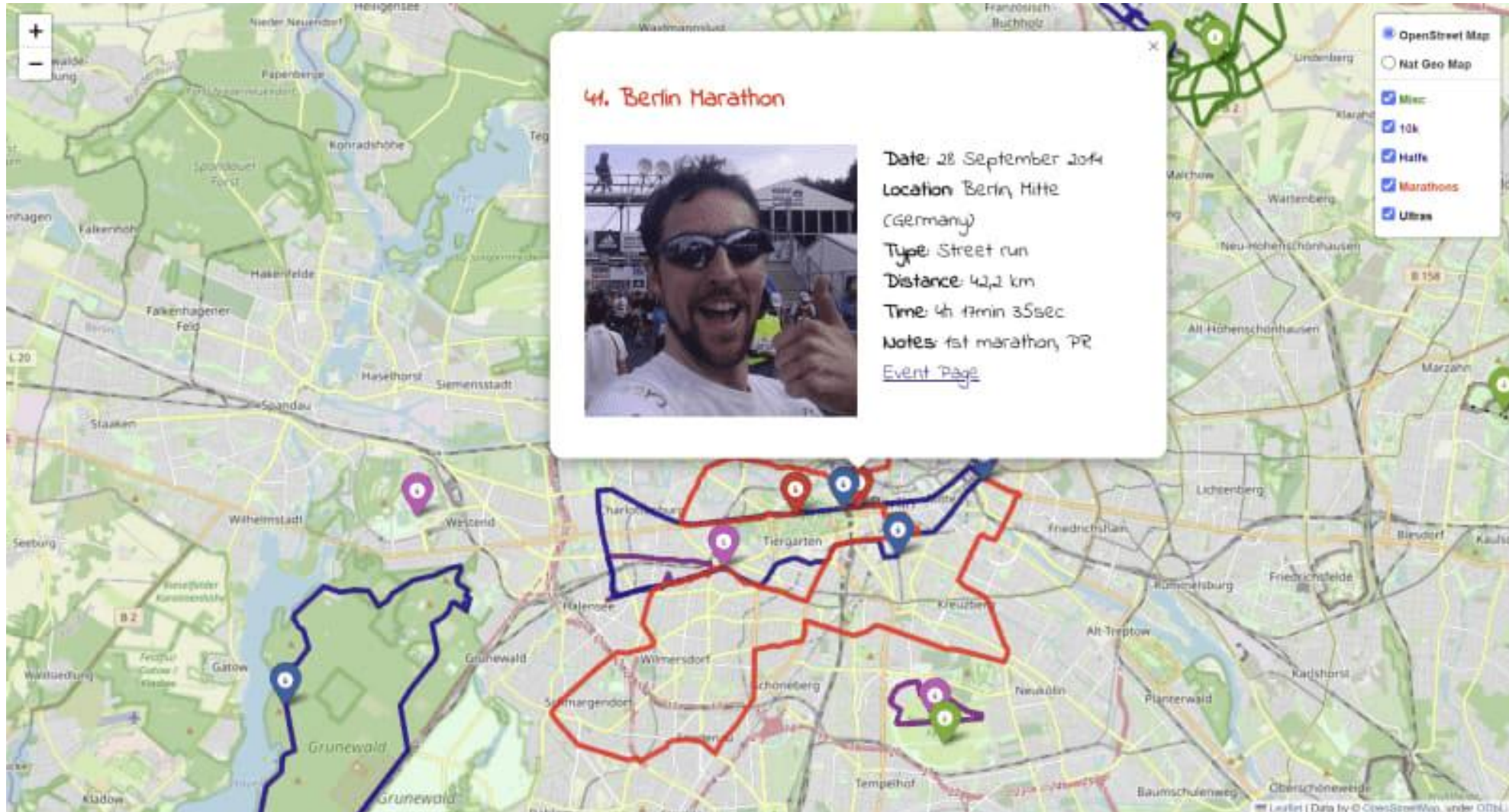


Animated Maps

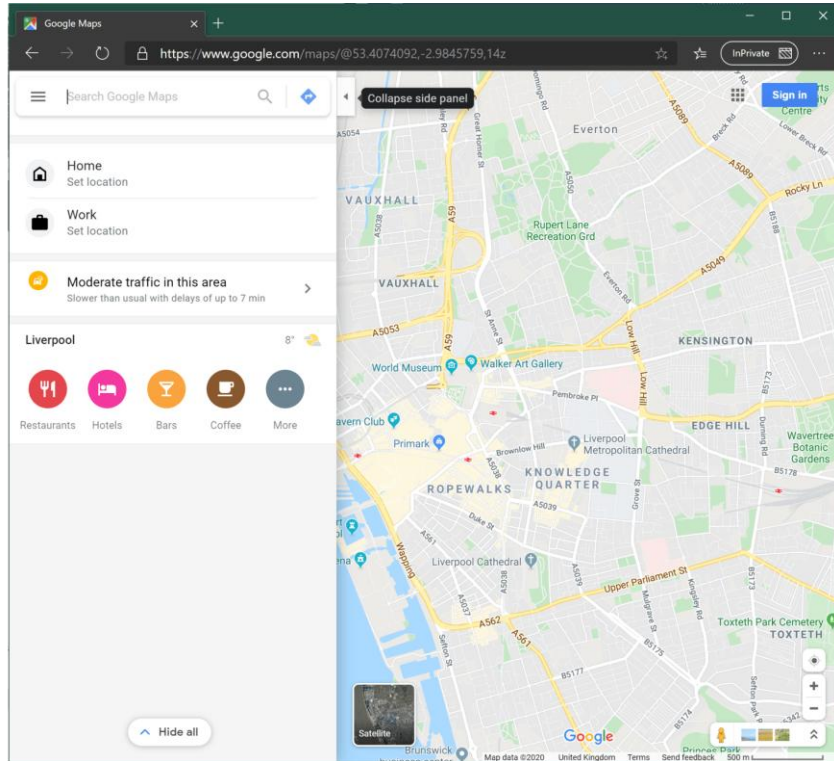


<https://www.maps.tnc.org/migrations-in-motion>

Interactive Maps



Web Maps



<https://globaldiplomacyindex.lowyinstitute.org>

The process of designing, implementing, generating, and delivering maps on the World Wide Web (Neumann 2008)

Building Blocks of a Map

Geographic Data

Data graphics visually display measured quantities by means of the combined use of points, lines, a coordinate system, numbers, symbols, words, shading, and color

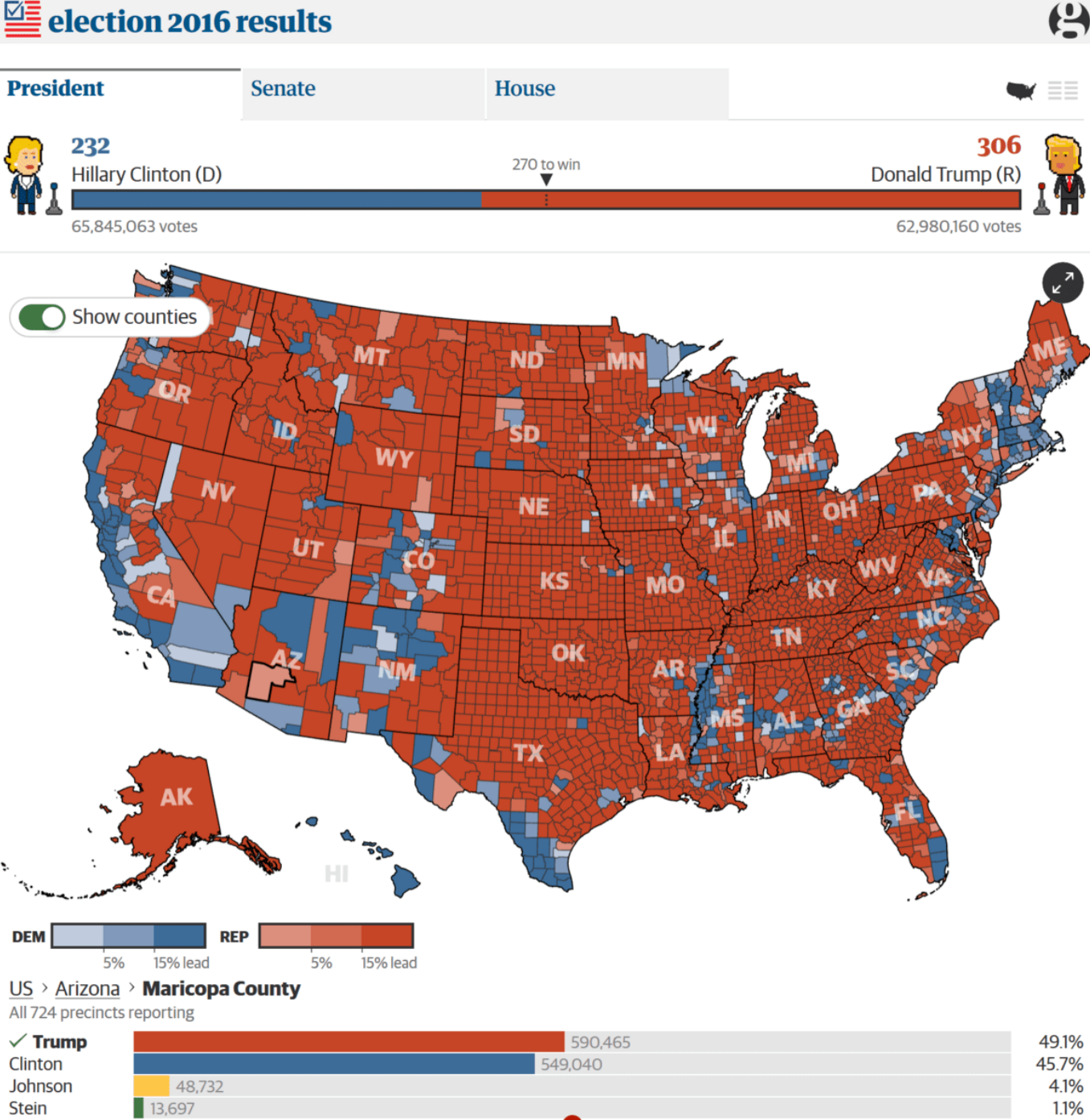
The Visual Display of Quantitative Information. Edward R. Tufte

Design

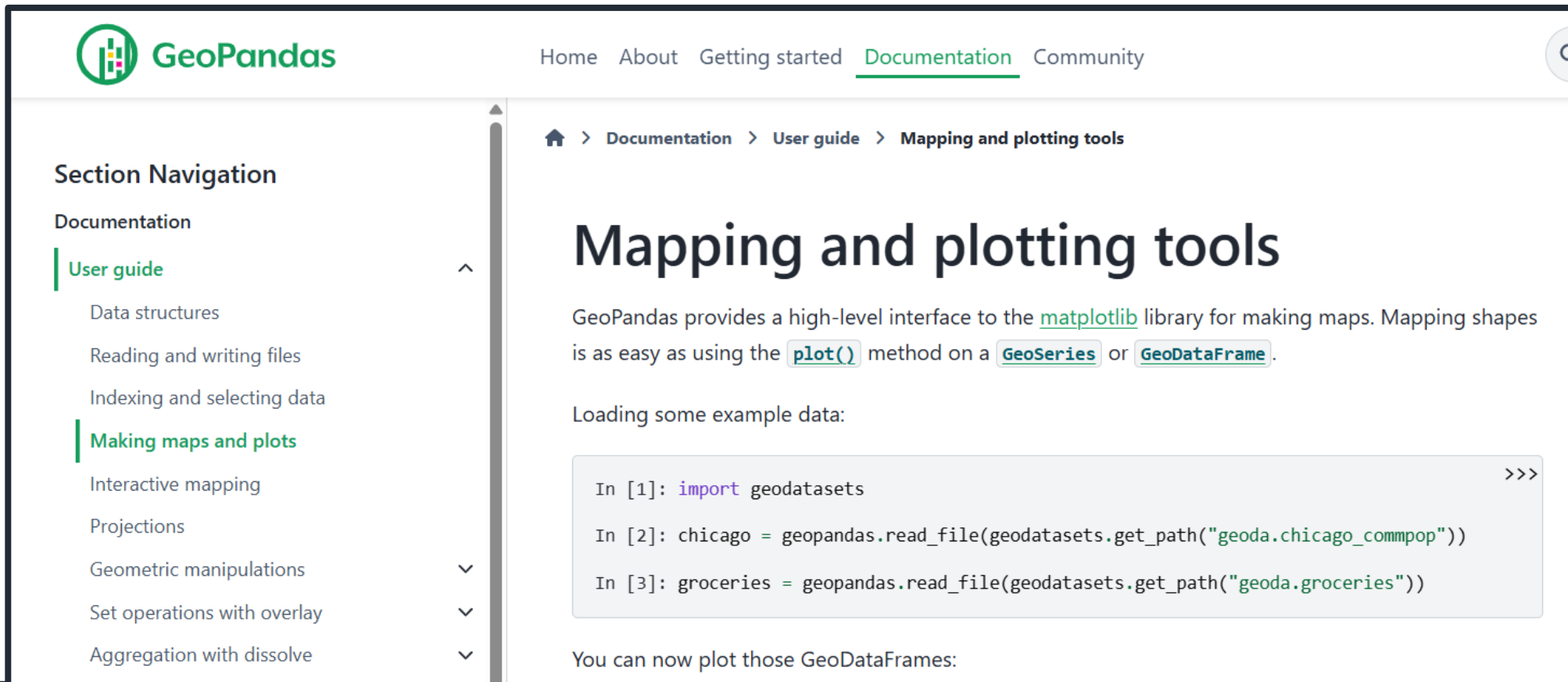
#4 For Those Of You Suffering From Election Fatigue. A Map Of Pubs In The Uk

Terrible Maps
9 June 2017 ·

For those of you suffering from election fatigue. A map of pubs in the UK



Software



The screenshot shows the GeoPandas documentation website. The top navigation bar includes links for Home, About, Getting started, Documentation (which is highlighted), and Community. The left sidebar, titled 'Section Navigation', lists various documentation topics, with 'User guide' and 'Making maps and plots' highlighted. The main content area is titled 'Mapping and plotting tools' and contains the following text:

GeoPandas provides a high-level interface to the [matplotlib](#) library for making maps. Mapping shapes is as easy as using the `plot()` method on a `GeoSeries` or `GeoDataFrame`.

Loading some example data:

```
In [1]: import geodatasets
In [2]: chicago = geopandas.read_file(geodatasets.get_path("geoda.chicago_commpop"))
In [3]: groceries = geopandas.read_file(geodatasets.get_path("geoda.groceries"))
```

You can now plot those GeoDataFrames:

QGIS

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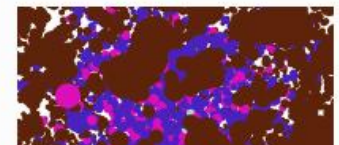
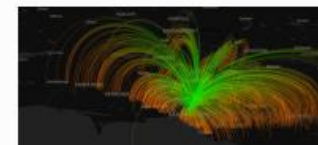
Spatial without Compromise



High-scale spatial rendering in Python, powered by [deck.gl](#).

Get started by [installing pydeck](#).

Gallery



Logistics

Material -> Webpage

<https://gdsl-ul.github.io/wma/>

Deployment ▾ Banja Kovi W Golan 2 E Etsy UI 5 G 108 pr 5 S Edit: \ 15 gds_py | g (58) Introd Build: 2 ds Best L 3 Geography panel 4 Is it po 2 uber 11 How to res Web M 7 Email 3 +

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https://gdsl-ul.github.io/wma/ Search Google

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Web Mapping and Geovisualisation

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WELCOME

Web Mapping and Geovisualisation

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WELCOME

This is the website for “Web Mapping and Geovisualisation” (module **ENVS456**) at the University of Liverpool. This course is designed and delivered by Dr. Gabriele Filomena and Dr. Elisabetta Pietrostefani from the Geographic Data Science Lab at the University of Liverpool, United Kingdom. The module has two main aims. It seeks to provide hands-on experience and training in:

- The design and generation of web-based mapping and geographical information tools.
- The use of software to access, analyse and visualize web-based geographical information.

The website is **free to use** and is licensed under the [Attribution-NonCommercial-NoDerivatives 4.0 International](#). A compilation of this web course is hosted as a GitHub repository that you can access:

- As an [html website](#).
- As a [GitHub repository](#).

CONTACT

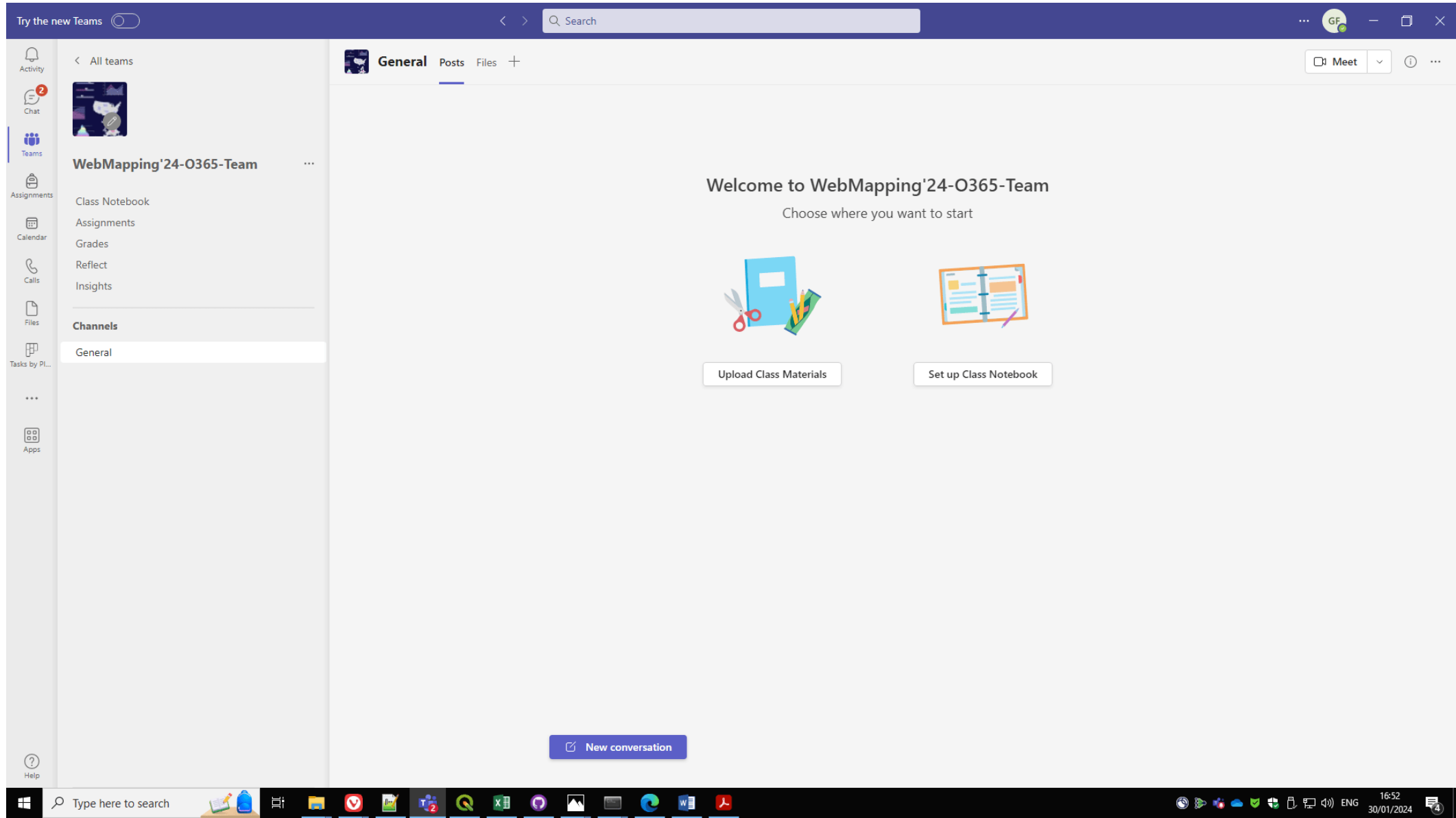
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Syllabus ->

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Support & Help -> Ms Teams



Course Principles & Philosophy

- Concepts : Lectures (website + slides)
 - Hands-on : Concepts in (interactive) action
 - Do-It-Yourself : practical material to do on your own
-
- This course is like a gym subscription
 - Principles over technology
 - Collaborate, do not copy --> PLAGIARISM